PROJECT REPORT

ON

“Smart Computer Aptitude and Skill Assessment System”

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CERTIFICATE

This is to certify that Mr. Sreenivas Bharadwaj (UID- 26MCA20094) has successfully completed the project titled **"Smart Computer Aptitude and Skill Assessment System"** at University Institute of Computing under my supervision and guidance in the fulfilment of requirements of first semester, Master of Computer Application- Specialization in General. Of Chandigarh University, Mohali, Punjab.

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ABSTRACT

Computer aptitude and technical skill assessment plays an important role in understanding a learner's current ability and identifying areas that require improvement. The project titled **"Smart Computer Aptitude and Skill Assessment System"** focuses on conducting a timed computer aptitude test, evaluating the answers automatically, and presenting the result through charts and graphs using web technologies.

The project uses HTML for the structure of the web pages, CSS for a responsive and professional user interface, and JavaScript for the test logic, timer, scoring and navigation. The Chart.js library is used to draw doughnut, radar, bar and line charts, and the browser's local storage is used to save previous attempts. The system maintains information such as question category, question text, answer options, correct answer, time spent per question, score, skill level and date of the attempt.

The system provides a Full Test of 20 questions, a Quick Test of 10 random questions, and category-wise practice in Logical Reasoning, Programming Basics, Networking & Web and Computer Fundamentals. A countdown timer, question navigator and keyboard shortcuts support the test. After submission, a dashboard displays the overall score, correct answers, time taken, skill level, answer breakdown, category-wise performance, time per question, personalised recommendations and a complete answer review. A history page shows the score trend across attempts.

The project demonstrates how HTML, CSS and JavaScript can be combined to build an interactive, data-driven assessment tool without a server. The current version uses a fixed question bank and a rule-based scoring method and is intended for educational demonstration rather than formal certification. The system can be further enhanced by integrating a database, secure user authentication, an administrator panel and adaptive question difficulty.

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INTRODUCTION

The Smart Computer Aptitude and Skill Assessment System is a web-based application developed to measure a user's computer aptitude and technical skills through a timed multiple-choice test. The main objective of this project is to provide a quick, interactive and visual way of evaluating basic computer knowledge and to help users identify the areas in which they need improvement.

The system is developed using HTML for the page structure, CSS for the responsive design, and JavaScript for the test logic, timer, scoring and navigation. The Chart.js library is used to display results as graphs, and the browser's local storage is used to keep a record of previous attempts. The test covers four categories: Logical Reasoning, Programming Basics, Networking & Web, and Computer Fundamentals.

After the test, the application calculates the score, classifies the user into a skill level (Beginner, Intermediate, Advanced or Expert) and displays a dashboard with charts, category-wise scores, time taken per question, personalised recommendations and a complete answer review. Overall, this project demonstrates how front-end technologies can be used to build an interactive assessment tool that gives instant and meaningful feedback.

PROBLEM STATEMENT

Conducting aptitude and skill tests manually or on paper can be time-consuming and may lead to errors in checking, totalling and analysing the results. Learners usually receive only a final mark and do not get clear feedback about their strong and weak areas, and it is difficult to track improvement over several attempts.

The Smart Computer Aptitude and Skill Assessment System aims to address these problems by providing an online platform that conducts the test, evaluates the answers instantly and presents the result visually. The system evaluates performance using the number of correct answers, category-wise scores and the time spent on each question. It generates a skill level and recommendations that help users understand their performance more effectively.

OBJECTIVES OF THE PROJECT

The main objective of the Smart Computer Aptitude and Skill Assessment System is to conduct computer aptitude tests efficiently and analyse the results visually.

The major objectives of the project are:

- **Online Assessment:** To conduct a timed multiple-choice test on computer aptitude and technical skills.
- **Instant Evaluation:** To calculate the score automatically and classify the user's skill level.
- **Category-wise Analysis:** To measure performance separately in four skill areas.
- **Visual Reports:** To present results using doughnut, radar, bar and line charts.
- **Time Management:** To provide a countdown timer and record the time spent on each question.
- **Personalised Feedback:** To recommend topics for the areas in which the score is below 60%.
- **Progress Tracking:** To save previous attempts in the browser and show the score trend.
- **User-friendly Interface:** To provide a responsive, professional design with light and dark themes.

Overall, the project aims to reduce manual effort, give instant feedback and support better understanding of a learner's computer skills.

SCOPE OF THE PROJECT

The Smart Computer Aptitude and Skill Assessment System aims to simplify skill testing and result analysis. It helps users check their computer knowledge and understand their strengths and weaknesses through a web-based platform.

The scope of the project includes:

- **Test Modes:** Full Test (20 questions), Quick Test (10 random questions) and practice in a single category.
- **Timer:** Selectable time limit of 5, 15 or 30 minutes, or no time limit.
- **Result Analysis:** Calculating score, accuracy, time taken and category-wise performance.
- **Skill Level:** Classifying users as Beginner, Intermediate, Advanced or Expert.
- **Recommendations:** Suggesting topics for improvement in weak areas.
- **History:** Saving previous attempts and showing the score trend.
- **Interface:** Responsive layout, light and dark themes and keyboard shortcuts.

The system can be further enhanced with a database, user login, an administrator panel for adding questions and adaptive difficulty levels.

SYSTEM REQUIREMENTS

1. Hardware Requirements

- Processor: Intel or AMD processor, or equivalent.
- RAM: Minimum 2 GB (4 GB recommended).
- Storage: At least 100 MB of free space.
- Input Devices: Keyboard and mouse.
- Output Device: Monitor or display.

2. Software Requirements

- Operating System: Windows 10/11, Linux, or macOS.
- Languages: HTML5, CSS3 and JavaScript (ES6).
- Charting Library: Chart.js 4.4.
- Code Editor: Visual Studio Code with the Live Server extension.
- Web Browser: Google Chrome, Microsoft Edge or Mozilla Firefox.
- Internet connection: Required to load Chart.js and the web font.

3. Technologies and Libraries

- **HTML:** Used to structure the pages, forms, buttons and result sections.
- **CSS:** Used to design the responsive interface, colours, cards and themes.
- **JavaScript:** Used for the timer, scoring, navigation, charts and local storage.
- **Chart.js:** Used to draw the doughnut, radar, bar and line charts.

TOOLS AND TECHNOLOGIES USED

The Smart Computer Aptitude and Skill Assessment System uses the following tools and technologies for development:

HTML: Used to create the structure of the web pages, including the home page, test screen, result dashboard, history and about pages.

CSS: Used to style the pages with a responsive grid layout, rounded cards, gradients and light and dark colour themes.

JavaScript: Used to load and shuffle questions, run the countdown timer, record answers, calculate the score and skill level, and switch between pages.

Chart.js: A JavaScript charting library used to draw the doughnut, radar, bar and line charts on the result and history pages.

Web Storage (localStorage): Used to save previous attempts in the browser so that the history page can show the score trend.

Google Fonts: The Plus Jakarta Sans font is used to give the interface a clean and professional look.

Visual Studio Code: Used to write, edit and manage the project code, together with the Live Server extension to run it locally.

Web Browser: Google Chrome or Microsoft Edge is used to access and test the application locally.

DATASET DESCRIPTION

The Smart Computer Aptitude and Skill Assessment System uses a question bank stored as a JavaScript array. The bank contains 20 multiple-choice questions, five in each of the four categories. The result of every attempt is stored in the browser's local storage.

The question bank includes the following attributes:

- **Category:** Logical Reasoning, Programming Basics, Networking & Web or Computer Fundamentals.
- **Question Text:** The question displayed to the user.
- **Options:** Four answer choices (A, B, C and D).
- **Correct Answer:** The index of the correct option used for evaluation.

Each saved attempt stores the following attributes:

- **User Name, Test Mode and Date:** Details of the attempt.
- **Score and Skill Level:** The percentage obtained and the level assigned.
- **Time per Question:** The seconds spent on each question during the attempt.

The system uses these attributes to calculate scores, classify skill levels, compare category-wise results and generate the charts and recommendations.

IMPLEMENTATION AND JAVASCRIPT CODE

The Smart Computer Aptitude and Skill Assessment System is implemented using HTML, CSS and JavaScript. HTML provides the pages and forms, CSS provides the design and themes, and JavaScript manages the test, the timer, the scoring and the charts. The application runs completely in the browser and does not need a server.

9.1 Implementation Steps

- **Interface Design:** Create the home, test, result, history and about pages using HTML and CSS.
- **Question Bank:** Prepare 20 questions in four categories as a JavaScript array.
- **Test Engine:** Add the countdown timer, question navigator, answer selection and keyboard shortcuts.
- **Result Processing:** Calculate score, category-wise percentages, time per question and skill level.
- **Charts:** Draw the doughnut, radar, bar and line charts using Chart.js.
- **History:** Save each attempt in local storage and display the score trend.
- **Testing:** Run the application in the browser and verify that every feature works correctly.

9.2 JavaScript Code

The following code demonstrates the basic logic used to calculate the score and classify the skill level.

```
// Question format: [category, question, options, correct option index]
const questions = [
  { q: "What is 25% of 480?", o: ["100","110","120","130"], a: 2 },
  { q: "Which memory is volatile?", o: ["ROM","SSD","RAM","Hard disk"], a: 2 }
];

function calculateResult(questions, answers) {
  let correct = 0;
  questions.forEach((q, i) => { if (answers[i] === q.a) correct++; });
  const percent = Math.round(correct / questions.length * 100);
  let level;
  if (percent >= 90) level = "Expert";
  else if (percent >= 70) level = "Advanced";
  else if (percent >= 40) level = "Intermediate";
  else level = "Beginner";
  console.log("Correct:", correct, "Score:", percent + "%", "Level:", level);
  return { correct, percent, level };
}

// Example: user answered both questions correctly
calculateResult(questions, [2, 2]);
```

OUTPUT AND SCREENSHOTS

1. Home Page

The home page asks for the user's name and lets the user choose the test mode (Full Test, Quick Test or single-category practice) and the time limit. Links to Home, History and About and a theme button are available at the top.

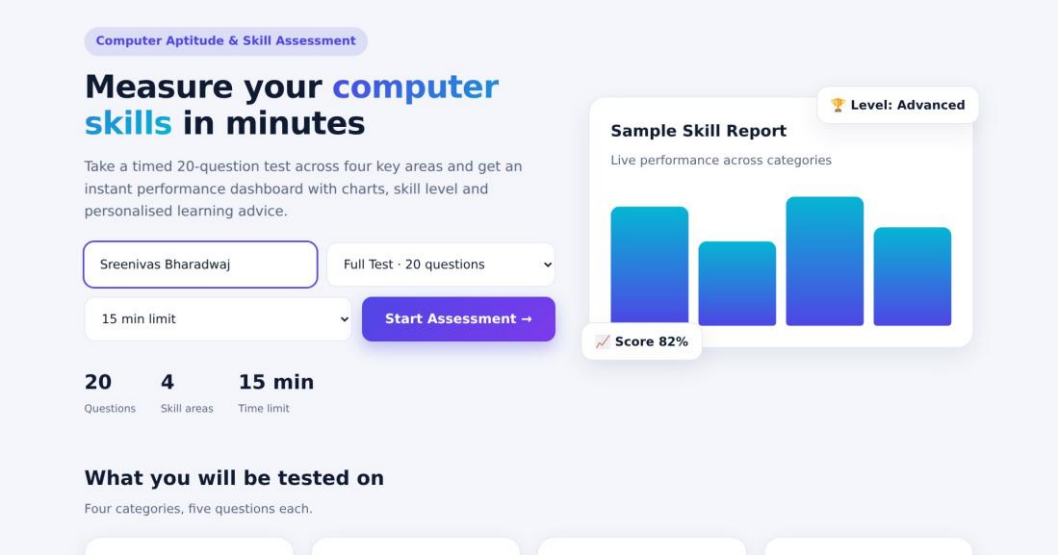


Figure 1: Home Page

2. Test Screen

The test screen shows the category, question number, countdown timer and progress bar. Numbered buttons let the user jump to any question, and Previous, Next and Exit test buttons control the test.

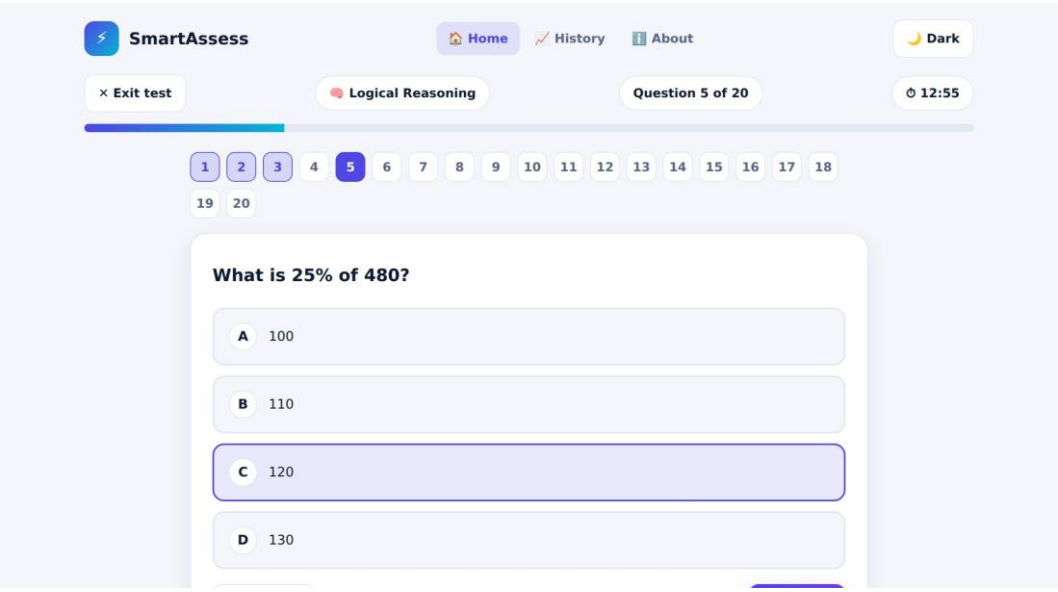


Figure 2: Test Screen

3. Results Dashboard

After submission the dashboard shows the overall score, correct answers, time taken, unanswered questions and the skill level, together with the answer breakdown doughnut chart and the skill radar chart.

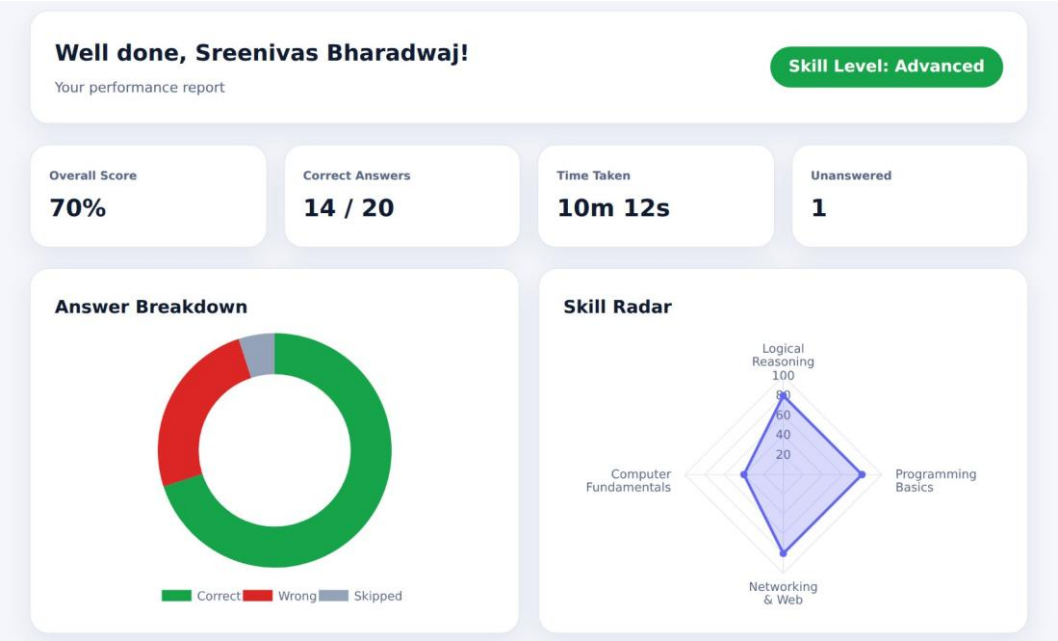


Figure 3: Results Dashboard

4. Detailed Analysis

The bar chart shows the time spent on each question (green for correct and red for wrong answers). Category score bars and recommendations show the areas that need improvement.

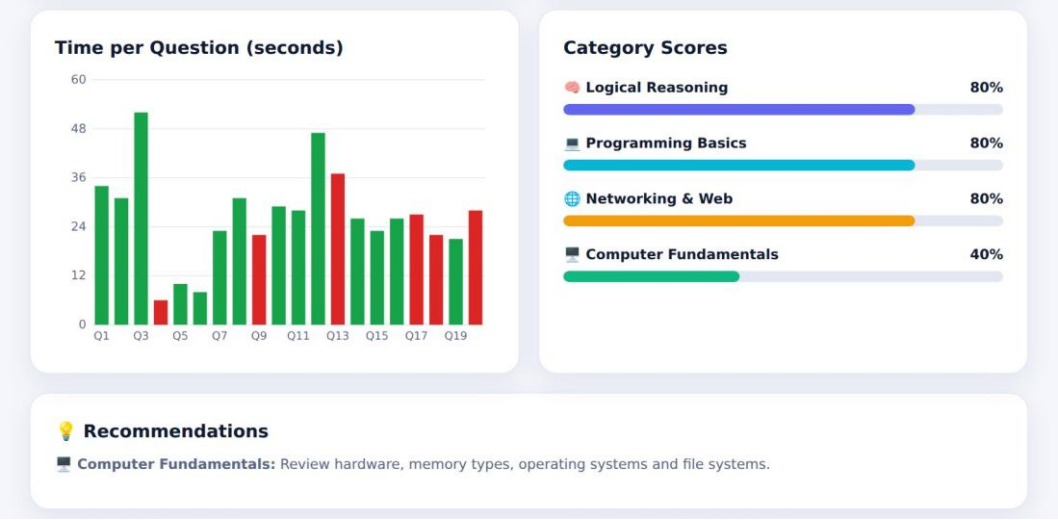


Figure 4: Detailed Analysis

5. Attempt History

The history page shows the number of attempts, best score and average score, a line chart of the score trend and the list of all saved attempts, with an option to clear the history.

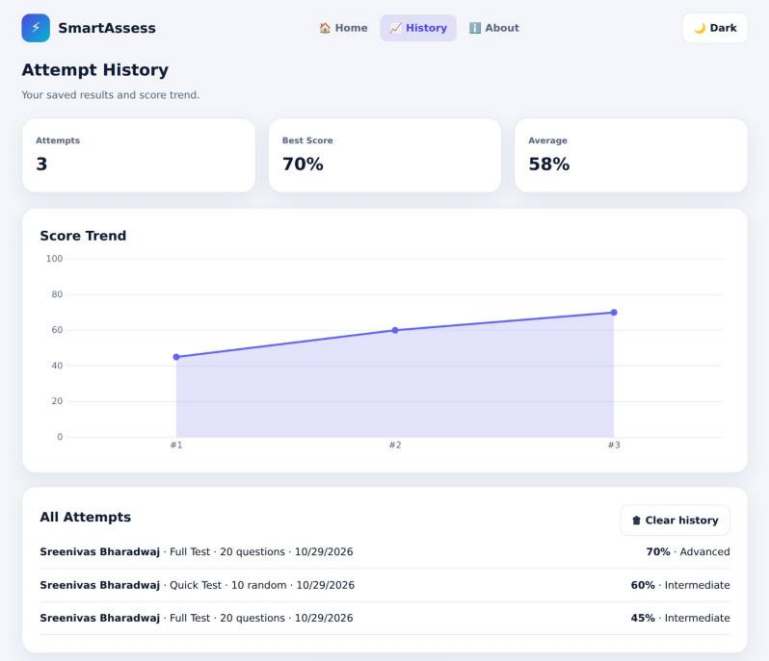


Figure 5: Attempt History

6. Dark Theme

The theme button switches the whole interface, including the charts, to a dark colour scheme that is comfortable to use in low light.

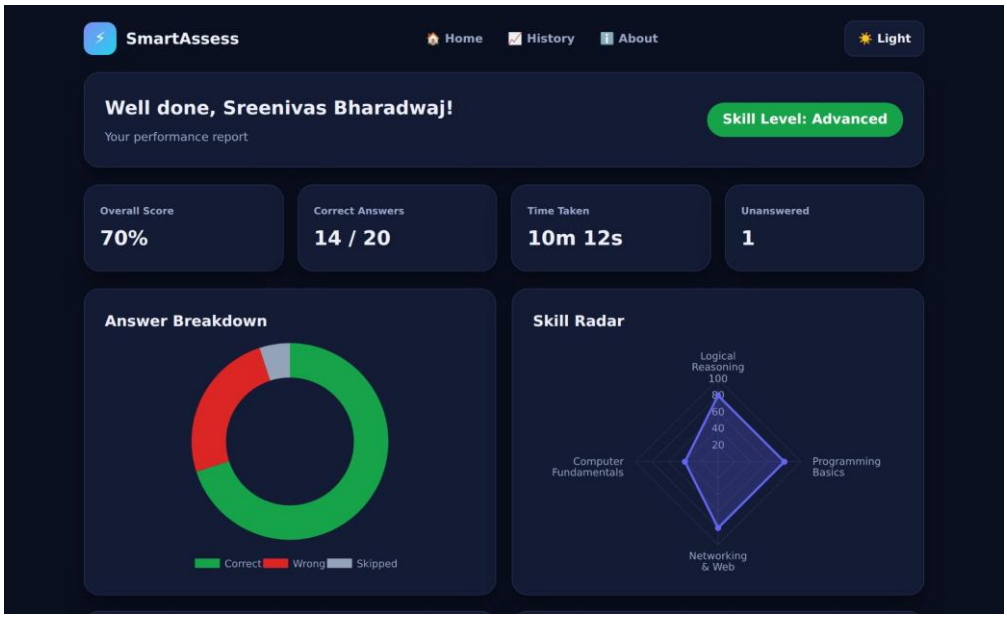


Figure 6: Dark Theme

SCORING MODEL

The score is calculated as the percentage of correct answers. Unanswered questions are counted as incorrect. Category-wise scores are calculated in the same way using only the questions of that category. The skill level is then assigned using the score ranges given below.

score (%) = (correct answers / total questions) * 100

Skill level	Rule
Expert	score ≥ 90 %
Advanced	score ≥ 70 % and < 90 %
Intermediate	score ≥ 40 % and < 70 %
Beginner	score < 40 %

Results :

The results of a sample Full Test attempt recorded during testing are shown below. The user answered 14 of the 20 questions correctly and one question was left unanswered, giving an overall score of 70 %, which is classified as Advanced. Computer Fundamentals is the weakest area (40 %), so the system recommends reviewing hardware, memory types, operating systems and file systems. Across the three sample attempts the history page shows the score rising from 45 % to 60 % and then 70 %.

Category	Questions	Correct	Score %	Remark
Logical Reasoning	5	4	80	Good
Programming Basics	5	4	80	Good
Networking & Web	5	4	80	Good
Computer Fundamentals	5	2	40	Needs improvement
Overall	20	14	70	Advanced

Testing

Each feature was exercised with valid, invalid and boundary inputs: starting the test with a blank name, leaving a test in progress (a second click is required to confirm), answering and moving between questions with the keyboard, switching the theme repeatedly, running a single-category practice test (five questions, radar chart hidden), automatic submission when the time limit ends, viewing the result page through the browser address, reloading the page to confirm that the history is retained, and clearing the history. The application handled each case with the expected message or behaviour in Google Chrome.

LIMITATIONS

The Smart Computer Aptitude and Skill Assessment System has certain limitations:

- **Fixed Question Bank:** The system contains only 20 questions, which are stored in the code and cannot be changed from the interface.
- **Basic Evaluation:** The system uses rule-based scoring and does not adapt question difficulty to the user's ability.
- **No Database or Login:** Results are saved only in the browser, so they are not shared between devices and are lost if the browser data is cleared.

- **Question Type:** Only multiple-choice questions are supported; coding and practical tasks are not included.
- **Internet Dependency:** Chart.js and the web font are loaded from the internet.
- **Not a Formal Certification:** The result is an indicative assessment for learning and practice.

FUTURE SCOPE

The Smart Computer Aptitude and Skill Assessment System can be improved in several ways in the future:

- **Database Integration:** Store questions and results on a server so that data is available from any device.
- **User Authentication:** Introduce secure login for students, teachers and administrators.
- **Administrator Panel:** Allow teachers to add, edit and delete questions and view all results.
- **Adaptive Testing:** Adjust the difficulty of questions according to the user's answers.
- **More Question Types:** Add coding questions, true/false, fill-in-the-blank and image-based questions.
- **Reports and Certificates:** Generate downloadable performance reports and certificates.
- **Machine Learning:** Use machine-learning techniques to predict skill growth and suggest personalised learning paths.

These enhancements can make the system more efficient, accessible and useful for educational institutions.

CONCLUSION

The Smart Computer Aptitude and Skill Assessment System provides an effective way to test computer aptitude and analyse the results using HTML, CSS and JavaScript. The system helps users take a timed test, calculate scores automatically, monitor their time, understand category-wise performance and identify the areas that need improvement.

The project demonstrates how web technologies can reduce manual effort and present assessment data through clear and attractive charts. Although the current version uses a fixed question bank and simple rule-based scoring, it provides a foundation for future improvements such as a database, user authentication, adaptive testing and online deployment. Overall, the project helps simplify skill assessment and supports a better understanding of a learner's computer skills.

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